

Day 7 — Eclipses: Why and When

Module: Moon Phases & Tithi · **Band:** Middle · **Time:** 45 min

Learning objective

Students will explain (with a diagram) why solar eclipses occur on *amāvasyā* and lunar eclipses on *pūrṇimā*, why eclipses do not happen every month, and what the safe ways to view a solar eclipse are.

Materials

- A small globe (or a large white ball — the Earth)
- A smaller dark ball (a tennis ball, painted dark — the Moon)
- A torch (the Sun)
- Two metres of string (to mark orbits)
- The activity-02 handout for the diagramming exercise
- Notebooks

Vocabulary introduced today

- *eclipse* — when one body's shadow falls on another (lunar eclipse: Earth's shadow on Moon) or one body passes between two others (solar eclipse: Moon between Sun and Earth).
- *ecliptic plane* — the plane of Earth's orbit around the Sun.
- *nodes* — the two points where the Moon's orbit crosses the ecliptic plane.

Lesson flow

Warm-up (5 min)

- Daily prompt: “*Did you see the Moon last night? What tithi?*” — 2 students share.
- “*You have a Moon diary now. Has any night's Moon been completely covered by Earth's shadow? Probably not. Why not? That's today.*”

Core (25 min)

Safety first (2 min, before anything else). Write on the board in capitals:

NEVER LOOK DIRECTLY AT A SOLAR ECLIPSE. Ordinary sunglasses do not protect your eyes. You need ISO 12312-2 certified eclipse glasses, or use pinhole projection.

Every student copies this into their workbook. We will revisit at end of class.

1. **The basic geometry — three bodies in a line.**

Set up the demo: torch (Sun) on one side, white ball (Earth) in middle, dark ball (Moon) on the other side. Show the two configurations:

- **Solar eclipse:** Sun — Moon — Earth (in that order). Moon casts a shadow on Earth. This can ONLY happen on *amāvasyā* (because that's the only time the Moon is between Sun and Earth).
- **Lunar eclipse:** Sun — Earth — Moon. Earth casts a shadow on Moon. This can ONLY happen on *pūrṇimā* (because that's the only time the Moon is opposite the Sun).

2. So why doesn't this happen every month?

“Every month has an amāvasyā and a pūrṇimā. If geometry were the only thing, we'd have a solar eclipse every new moon and a lunar eclipse every full moon. Clearly we don't.”

The reason: **the Moon's orbit is tilted by $\sim 5^\circ$ relative to Earth's orbit around the Sun.** Show this with the string and the two balls — Earth's orbit is one flat ring; the Moon's orbit is a smaller ring tilted by 5° .

Most months, when the Moon is at *amāvasyā* (between Earth and Sun), it's either above or below the Sun-Earth line — its shadow misses Earth. Same for *pūrṇimā* — Earth's shadow misses the Moon.

An eclipse only happens when the Moon is at *amāvasyā* OR *pūrṇimā* AND it's near one of the two nodes (the points where the Moon's tilted orbit crosses the flat ecliptic plane). That's why eclipses come in pairs (a solar and a lunar within ~ 14 days) and in “eclipse seasons” twice a year.

3. How many eclipses per year?

- Solar: 2 to 5 per year, somewhere on Earth.
- Lunar: 0 to 3 per year.
- From any one location, a total solar eclipse is rare — the path of totality is narrow, and most observers see only partial eclipses.

4. Solar vs. lunar — what they look like.

- **Solar eclipse:** the Sun briefly turns into a “ring of fire” (annular) or vanishes completely (total). Daylight goes blue-grey. Birds may roost. Only visible from the narrow shadow strip.
- **Lunar eclipse:** the Moon turns dim coppery-red (“blood moon”) because the only sunlight reaching it is sunlight bent through Earth's atmosphere. Visible from the entire night-side of Earth. Lasts up to ~ 100 minutes.

5. Indian astronomical sources knew this.

The *Sūrya-siddhānta* (c. 5th–10th c. CE) gives algorithms for eclipse prediction — start time, duration, magnitude. Āryabhaṭa (5th c. CE) had a clear understanding of eclipse geometry (Moon's shadow, Earth's shadow, ecliptic plane) and rejected mythological explanations (the “Rāhu swallowing the Sun”

story is older and pre-scientific; classical Indian astronomers had a separate, mathematical account).

Activity (10 min) — Diagram the geometry

This is `activities/activity-02-eclipse-geometry.md` in full. Students:

1. Receive the activity sheet (with two unfinished diagrams — Sun in one position, Earth in another, Moon's orbit drawn around Earth).
2. Place the Moon (a sticker or a drawn circle) at the eclipse positions: between Sun and Earth (solar) and opposite (lunar).
3. Draw the shadow cones.
4. Mark the two nodes on the Moon's orbit and shade the regions where an eclipse is possible.
5. Below the diagram, in 3 sentences, explain: *why does an eclipse not happen every amāvasyā / pūrṇimā?*

Wrap-up (5 min)

- Read the safety rule aloud as a class. Once more.
- Ask: “*What’s the safest way to watch a solar eclipse?*” (Eclipse glasses with ISO certification; or pinhole projection.) “*What about a lunar eclipse?*” (Look directly — no danger.)
- **Preview Day 8:** “*Tomorrow you start your project.*”

Student-facing content

Solar eclipse: Sun, Moon, Earth in line (Moon in middle). Happens on *amāvasyā*. Moon's shadow falls on Earth.

Lunar eclipse: Sun, Earth, Moon in line (Earth in middle). Happens on *pūrṇimā*. Earth's shadow falls on Moon.

Why not every month? The Moon's orbit is tilted $\sim 5^\circ$ from Earth's orbit around the Sun. So usually the Moon's shadow misses Earth (at *amāvasyā*) or Earth's shadow misses the Moon (at *pūrṇimā*). Eclipses happen only when the Moon is *both* at *amāvasyā/pūrṇimā* *and* close to one of the two nodes (where the two orbital planes cross).

⚠️ **NEVER look directly at a solar eclipse.** Use ISO 12312-2 certified eclipse glasses, or project the Sun through a pinhole onto paper. Sunglasses, smoked glass, photographic film do not work. Lunar eclipses are safe to view directly.

Homework

- Continue the Moon diary.
- Look up: what is the date of the next solar eclipse and the next lunar eclipse visible from India? Note in your workbook.

Differentiation

- **One tier up:** Find the dates of the nodes for this year. (The two nodes drift backward through the ecliptic — they complete one full circuit in ~18.6 years. This is the *Saros cycle*'s underlying mechanism.) Why does this give us “eclipse seasons” twice a year?
- **One tier down:** Skip the node-drift detail. Focus on: solar eclipse = amāvasyā + alignment, lunar eclipse = pūrṇimā + alignment. The “tilted orbit” is the key idea; the nodes are the precise version of it.

Teacher notes

- This is the single highest-safety-risk day in the entire IKS curriculum if there's a solar eclipse during the school term. Even if there isn't one this term, drill the safety rule. Eclipses come around.
- Some students will know the *Rāhu* story (Rāhu the demon swallowing the Sun/Moon during eclipses). Acknowledge it as a pre-scientific narrative shared by many cultures, then return to the geometric explanation. Classical Indian astronomers like Āryabhaṭa explicitly rejected the *Rāhu* explanation in favour of the shadow-geometry account.
- The “5° tilt” number is the key. Repeat it. Without it, the “why not every month” question has no answer.

Citation(s) used in this lesson

- Paraphrase from *Sūrya-siddhānta*, chapters 4–5 (computation of solar and lunar eclipses): the text gives detailed algorithms for predicting eclipse timing using the geometry of the Moon's nodes. Cited by text and chapter range only.
- Modern reference: NASA's eclipse pages (search “NASA Eclipse Website” — official, free).